

Glumet XR Tablet

TABLET DESCRIPTION

Glumet XR Tablet 500 mg

White to off white, oblong biconvex tablet with Pharmaniaga icon on one side and plain on the other.

Glumet XR Tablet 750 mg

White to off white, oblong biconvex tablet and plain on both sides.

COMPOSITION

Glumet XR Tablet 500 mg

Each tablet contains Metformin hydrochloride 500 mg

Glumet XR Tablet 750 mg

Each tablet contains Metformin hydrochloride 750 mg

ACTIONS

Pharmacodynamics

ORAL ANTI-DIABETICS

(A10BA02: Gastrointestinal tract and metabolism)

Mechanism of action

Metformin is a biguanide with antihyperglycaemic effects, lowering both basal and postprandial hyperglycaemia. It does not stimulate insulin secretion and therefore does not cause hypoglycaemia. Metformin reduces basal hyperinsulinemia, and in combination with insulin, reduces insulin requirement.

Metformin exerts its antihyperglycaemic effect via multiple mechanisms:

1. Metformin reduces hepatic glucose production.
2. Metformin facilitates peripheral glucose uptake and utilization, in part by increasing insulin action.
3. Metformin alters glucose turnover in the gut: Uptake from circulation is increased and absorption from food is decreased. Additional mechanisms attributed to the gut include an increase in release of glucagon-like peptide 1 (GLP-1) and a decrease of bile acid resorption. Metformin alters the gut microbiome.

Metformin can improve the lipid profile in hyperlipidaemic individuals.

In published clinical studies, use of metformin was associated with either a stable body weight or modest weight loss.

Metformin is an adenosine monophosphate-protein-kinase (AMPK) activator and increases the transport capacity of all types of membrane glucose transporters (GLUTs).

Clinical efficacy

Treatment of type 2 diabetes mellitus

The prospective randomised (UKPDS) study has established the long-term benefit of intensive blood glucose control in overweight type 2 diabetic patients treated with immediate release metformin as first-line therapy after diet failure. Analysis of the results for overweight patients treated with metformin after failure of diet alone showed a significant reduction of the absolute risk for:

- diabetes-related complication in the metformin group (29.8 events/1000 patient-years) versus diet alone (43.3 events/1000 patient-years), and versus the combined sulfonylurea and insulin monotherapy groups (40.1 events/1000 patient-years);
- diabetes-related mortality: metformin (7.5 events/1000 patient-years), diet alone (12.7 events/1000 patient-years);
- overall mortality: metformin (13.5 events/1000 patient-years) versus diet alone (20.6 events/1000 patient-years), and versus the combined sulfonylurea and insulin monotherapy groups (18.9 events/1000 patient-years);
- myocardial infarction: metformin (11 events/1000 patient-years), diet alone (18 events/1000 patient-years).

Benefit regarding the outcome has not been shown for metformin used as second-line therapy, in combination with a sulfonylurea.

In type 1 diabetes, the combination of metformin and insulin has been used in selected patients, but the clinical benefit of this combination has not been formally established.

Pharmacokinetics

Absorption

Following a single oral administration of 1500 mg of Glumet XR 750 mg, a mean peak plasma concentration of 1193 ng/ml is achieved with a median value of 5 hours and a range of 4 to 12 hours. Metformin absorption is significantly delayed compared to the immediate release tablet with a Tmax at 7 hours (Tmax for the immediate release tablet is 2.5 hours).

At steady state, similar to the immediate-release formulation, Cmax and AUC are not proportionally increased to the administered dose. The AUC after a single oral administration of 2000 mg of metformin extended release tablet is similar to that observed after administration of 1000 mg of Metformin immediate-release tablets b.i.d.

Intrasubject variability of Cmax and AUC of Metformin extended release is comparable to that observed with metformin immediate-release tablets. When the extended release tablet is administered in fasting conditions the AUC is decreased by 30% (both Cmax and Tmax are unaffected).

Metformin absorption from the extended release formulation is not altered by meal composition. No accumulation is observed after repeated administration of up to 2000 mg of metformin as extended release tablets.

Distribution

Plasma protein binding is negligible. Metformin partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean Vd ranged between 63-276 L.

Metabolism

Metformin is excreted unchanged in the urine. No metabolites have been identified in humans.

Elimination

Renal clearance of metformin is > 400 ml/min, indicating that metformin is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 hours. When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life prolonged, leading to increased levels if metformin in plasma.

Characteristics in specific groups of patients

Renal impairment

The available data in subjects with moderate renal insufficiency are scarce and no reliable estimation of the systemic exposure to metformin in this subgroup as compared to subjects with normal renal function could be made. Therefore, the dose adaptation should be made upon clinical efficacy/tolerability considerations.

Preclinical safety data

Published preclinical data reveal no special hazard for humans based on conventional studies on safety, pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential, and reproductive toxicity.

INDICATION

Reduction in the risk or delay of the onset of type 2 diabetes mellitus in adult, overweight patients with IGT* and/or IFG*, and/or increased HbA1C who are:

- at high risk for developing overt type 2 diabetes mellitus and
- still progressing towards type 2 diabetes mellitus despite implementation of intensive lifestyle change for 3 to 6 months

Treatment with Glumet XR must be based on a risk score incorporating appropriate measures of glycaemic control and including evidence of high cardiovascular risk.

Lifestyle modifications should be continued when metformin is initiated, unless the patient is unable to do so because of medical reasons.

*IGT: Impaired Glucose Tolerance; IFG: Impaired Fasting Glucose

Treatment of type 2 diabetes mellitus in adults, when dietary management and exercise alone does not result in adequate glycaemic control. Glumet XR may be used as monotherapy or in combination with other oral antidiabetic agents, or with insulin.

A reduction of diabetic complications has been shown in overweight type 2 diabetic patients treated with immediate release metformin as first-line therapy after diet failure.

CONTRAINDICATIONS

- Hypersensitive to metformin hydrochloride or any of the excipients.
- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis). Diabetic ketoacidosis should be treated with insulin.
- Diabetic pre-coma.
- Severely reduced kidney function (GFR <30 mL/min).
- Acute conditions with the potential to alter renal function such as:
 - Dehydration,
 - Severe infection,
 - Shock.
- Disease which may cause tissue hypoxia (especially acute disease, or worsening of chronic disease) such as:

- Decompensated heart failure,
- Respiratory failure,
- Recent myocardial infarction,
- Shock.
- Hepatic insufficiency, acute alcohol intoxication, alcoholism.

DOSAGE AND ADMINISTRATION

Renal impairment

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

GFR mL/min	Total maximum daily dose (to be divided into considerations 2-3 daily doses) *	Additional considerations
60-89	3000 mg	Dose reduction may be considered in relation to declining renal function.
45-59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of Metformin. The starting dose is at most half of the maximum dose.
30-44	1000 mg	
<30	-	Metformin is contraindicated.

* The text "to be divided into 2-3 daily doses" should be omitted for extended release products containing metformin as single agent.

Monotherapy and combination with other oral antidiabetic agents:

For Glumet XR 500mg

- The usual starting dose of Glumet XR 500mg is one tablet daily with the evening meal.
- After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastro-intestinal tolerability. The maximum recommended dose is 4 tablets of Glumet XR 500mg daily with the evening meal.
- Dosage increases should be made in increments of 500mg every 10-15 days, up to a maximum of 2000mg once daily with the evening meal.

For Glumet XR 750mg

- The usual starting dose of Glumet XR 750mg is one tablet daily with the evening meal.
- After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastro-intestinal tolerability. The maximum recommended dose is 2 tablets of Glumet XR 750mg once daily with the evening meal.

In patients already treated with metformin tablets, the starting dose of Glumet XR should be equivalent to the daily dose of metformin immediate release tablets.

If transfer from another oral antidiabetic agent is intended: discontinue the other agent and initiate Glumet XR at the dose indicated above.

Combination with insulin: Metformin and insulin may be used in combination therapy to achieve better blood glucose control. The usual starting dose of Glumet XR is one tablet of 500mg once daily with the evening meal, or one tablet of 750mg once daily up to a maximum of 1500mg (for Glumet XR 750mg) and 2000mg (for Glumet XR 500mg) with the evening meal, while insulin dosage is adjusted on the basis of blood glucose measurement.

Elderly: Due to the potential for decreased renal function in elderly subjects, the metformin dosage should be adjusted based on renal function. Regular assessment of renal function is necessary.

Benefit in the reduction of risk or delay of the onset of type 2 diabetes mellitus has not been established in patients 75 years and older and metformin initiation is therefore not recommended in these patients.

Children: In the absence of available data. Glumet XR should not be used in children.

Posology and method of administration

Adults with normal renal function (GFR ≥ 90 mL/min)

Reduction in the risk or delay of the onset of type 2 diabetes

- Metformin should only be considered where intensive lifestyle modifications for 3 to 6 months have not resulted in adequate glycaemic control.
- The therapy should be initiated with one tablet Glumet XR 500mg once daily with the evening meal.

- After 10 to 15 days dose adjustment on the basis of blood glucose measurements is recommended (OGTT and/or FPG and/or HbA1c values to be within the normal range). A slow increase of dose may improve gastro-intestinal tolerability. The maximum recommended dose of Glumet XR 500mg is 4 tablets (2000mg) once daily with the evening meal or of Glumet XR 750mg is 2 tablets (1500mg) once daily with the evening meal.
- It is recommended to regularly monitor (every 3-6 months) the glycaemic status (OGTT and/or FPG and/or HbA1c value) as well as the risk factors to evaluate whether treatment needs to be continued, modified or discontinued.
- A decision to re-evaluate therapy is also required if the patient subsequently implements improvements to diet and/or exercise, or if changes to the medical condition will allow increased lifestyle interventions to be possible.

ROUTE OF ADMINISTRATION

Oral

WARNINGS AND PRECAUTIONS

Lactic acidosis

Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected

symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal function:

GFR should be assessed before treatment initiation and regularly there after. Metformin is contraindicated in patients with GFR <30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function.

Administration of iodinated contrast agent:

Intravascular administration of iodinated contrast agents may lead to contrast induced nephropathy, resulting in metformin accumulation and an increased risk of lactic acidosis. Metformin should be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable.

Surgery:

Metformin must be discontinued at the time of surgery with general, spinal or epidural anaesthesia. Therapy may be restarted no earlier than 48 hours following surgery or resumption of oral nutrition and provided that renal function has been re-evaluated and found to be stable.

Cardiac function:

Patients with heart failure are more at risk of hypoxia and renal insufficiency. In patients with stable chronic heart failure, metformin may be used with a regular monitoring of cardiac and renal function. For patients with acute and unstable heart failure, metformin is contraindicated.

Elderly:

Due to limited therapeutic efficiency data in the reduction of risk or delay of type 2 diabetes in patients 75 years and older, metformin initiation is not recommended in these patients.

Other precautions:

- Metformin may reduce vitamin B12 serum levels. The risk of low vitamin B12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic

vitamin B12 monitoring could be necessary in patients with risk factors for vitamin B12 deficiency. Metformin therapy should be continued for as long as it is tolerated and not contraindicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines.

- The tablet shells may be present in the faeces. Patients should be advised that this is normal.
- Metformin alone never causes hypoglycaemia, although caution is advised when it is used in combination with insulin or other oral antidiabetics (e.g. sulphonylureas or meglitinides).
- All patients should continue their diet with regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.
- The usual laboratory tests for diabetes monitoring should be performed regularly.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Glumet XR monotherapy does not cause hypoglycaemia and therefore has no effect on the ability to drive or to use machines.

However, patients should be alerted to the risk of hypoglycaemia when metformin is used in combination with other antidiabetic agents (sulphonylureas, insulin, meglitinides).

PREGNANCY AND LACTATION

Pregnancy

Uncontrolled diabetes during pregnancy (gestational or permanent) is associated with increased risk of congenital abnormalities and perinatal mortality. When the patient plans to become pregnant and during pregnancy, it is recommended that impaired glycaemic control or diabetes are not treated with metformin. For diabetes it is recommended that insulin should be used to maintain blood glucose levels as close to normal as possible to reduce the risk of malformations of the foetus.

Fertility

Fertility of male or female rats was unaffected by metformin when administered at doses as high as 600 mg/kg/day, which is approximately three times the maximum recommended human daily dose based on body surface area comparisons.

Lactation

Metformin is excreted into human breast milk. No adverse effects were observed in breastfed newborns/infants. However, as only limited data are available, breastfeeding is not recommended during

metformin treatment. A decision on whether to discontinue breastfeeding should be made, taking into account the benefit of breastfeeding and the potential risk to adverse effect on the child.

DRUG INTERACTIONS

Concomitant use is not recommended

Alcohol

Increased risk of lactic acidosis in acute alcohol intoxication, particularly in case of:

- Fasting or malnutrition,
- Hepatic insufficiency.

Avoid consumption of alcohol and alcohol-containing medications.

Iodinated contrast agents

Intravascular administration of iodinated contrast agents may lead to contrast induced nephropathy, resulting in metformin accumulation and an increased risk of lactic acidosis. Metformin should be discontinued prior to or at the time of the imaging procedure and not restarted until at least 48 hours after, provided that renal function has been re-evaluated and found to be stable.

Combinations requiring precautions for use

Some medicinal products can adversely affect renal function which may increase the risk of lactic acidosis, e.g. NSAIDs, including selective cyclo-oxygenase (COX) II inhibitors, ACE inhibitors, angiotensin II receptor antagonists and diuretics, especially loop diuretics. When starting or using such products in combination with metformin, close monitoring of renal function is necessary.

Medicinal products with intrinsic hyperglycaemic activity (e.g. glucocorticoids (systemic and local routes) and sympathomimetics).

More frequent blood glucose monitoring may be required, especially at the beginning of treatment. If necessary, adjust the metformin dosage during therapy with the other drug and upon its discontinuation.

Metformin is a substrate of both organic cation transporters (OCT) 1 and OCT2. Co-administration of metformin with:

- Inhibitors of OCT1 (such as verapamil) may reduce efficiency of metformin.
- Inducers of OCT1 (such as rifampicin) may increase gastrointestinal absorption and efficacy of metformin.
- Inhibitors of OCT2 (such as cimetidine, dolutegravir, ranolazine, trimethoprim, vandetanib, isavuconazole) may decrease the

renal elimination of metformin and thus lead to an increase in metformin plasma concentrations.

- Inhibitors of both OCT1 and OCT2 (such as crizotinib, olaparib) may alter efficacy and renal elimination of metformin.

Caution is therefore advised, especially in patients with renal impairment, when these drugs are co-administered with metformin, as metformin plasma concentration may increase. If needed, dose adjustment of metformin may be considered as OCT inhibitors/inducers may alter the efficiency of metformin.

ADVERSE REACTIONS

In post marketing data and in controlled clinical studies, adverse event reporting in patients treated with Glumet XR was similar in nature and severity to that reported in patients treated with Metformin immediate-release.

During treatment initiation, the most common adverse reactions are nausea, vomiting, diarrhoea, abdominal pain and loss of appetite, which resolve spontaneously in most cases.

The following undesirable effects may occur with metformin.

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Frequencies are defined as follows: very common: $\geq 1/10$; common $\geq 1/100$, $< 1/10$; uncommon $\geq 1/1,000$, $< 1/100$; rare $\geq 1/10,000$, $< 1/1,000$; very rare $< 1/10,000$.

Metabolism and nutritional disorders

Very rare: Lactic acidosis

Common: Vitamin B12 decrease/ deficiency

Nervous system disorder

Common: Taste disturbance

Gastrointestinal disorders

Very common: gastrointestinal disorders such as nausea, vomiting, diarrhoea, abdominal pain, and loss of appetite.

These undesirable effects occur most frequently during initiation of therapy and resolve spontaneously in most cases. A slow increase of the dose may also improve gastrointestinal tolerability.

Hepatobiliary disorders

Very rare: Isolated reports of liver function test abnormalities or hepatitis resolving upon metformin discontinuation.

Skin and subcutaneous tissue disorders

Very rare: Skin reactions such as erythema, pruritus, urticaria.

OVERDOSAGE

Hypoglycaemia has not been seen with metformin doses of up to 85 g, although lactic acidosis has not occurred in such circumstances. High overdose or concomitant risk of metformin may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in hospital. The most effective method to remove lactate and metformin is haemodialysis.

STORAGE CONDITION

Store below 30°C.

SHELF-LIFE

Products should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

Glumet XR Tablet 500 mg and 750 mg

In blister of 10x10 tablets.

DATE OF REVISION

16th January 2025

PRODUCT REGISTRATION HOLDER /

MANUFACTURER:

PHARMANIAGA MANUFACTURING BERHAD (198001006232)

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